

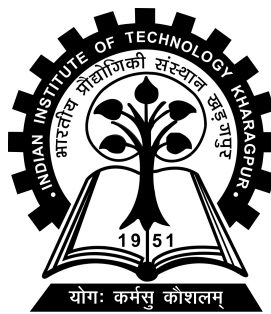
Learning correspondence estimation for deformable 3D shapes

Project-II (CS47006) report submitted to
Indian Institute of Technology Kharagpur
in partial fulfilment for the award of the degree of
Integrated Dual Degree (B.Tech + M.Tech)
in
Computer Science and Engineering

by

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Spring Semester, 2025-26

April 10, 2026

DECLARATION

I certify that

- (a) The work contained in this report has been done by me under the guidance of my supervisor.
- (b) The work has not been submitted to any other Institute for any degree or diploma.
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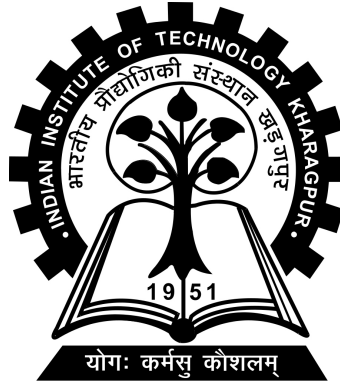
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CERTIFICATE

This is to certify that the project report entitled “Learning correspondence estimation for deformable 3D shapes” submitted by Lubesh Kumar Sharma (Roll No. 22CS30065) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Integrated Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering is a record of bona fide work carried out by him under my supervision and guidance during Spring Semester, 2025-26.

Date: April 10, 2026

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Abstract

Name of the student: **Lubesh Kumar Sharma**

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Degree for which submitted: **Integrated Dual Degree (B.Tech + M.Tech)**

Department: **Department of Computer Science and Engineering**

Thesis title: **Learning correspondence estimation for deformable 3D shapes**

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Estimating dense point-wise correspondences between non-rigidly deforming 3D shapes is a fundamental task in computer vision and geometry processing. While classical methods often rely on well-defined manifold mesh connectivity, this thesis explores a self-supervised framework designed to operate directly on unstructured point clouds, providing a more versatile solution for raw geometric data. The proposed approach utilizes an Adaptive Siamese Masked Autoencoder (ASMAE) architecture to learn robust geometric representations without manual supervision.

Heat Kernel Signatures (HKS) are employed as intrinsic descriptors, extracted from localized point-based graphs to capture the structural properties of the shapes. To enhance feature learning, an adaptive adversarial masking strategy is utilized to hide salient geometric regions, compelling the model to reconstruct the missing information by leveraging context from a target reference shape through cross-attention mechanisms. The framework is implemented within a student-teacher learning

paradigm, ensuring feature consistency and increased resilience to data perturbations and noise. To address the inherent challenges of bilateral symmetry and multiple-to-one vertex mappings, a global optimization module based on Optimal Transport and the Sinkhorn algorithm is integrated into the training process. Experimental evaluations conducted on standard datasets, including FAUST, SCAPE, SHREC'19, and SURREAL, demonstrate that the integrated pipeline produces consistent correspondence results and offers a robust strategy for unsupervised non-rigid matching directly from raw point cloud inputs.

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Abbreviations

3D	Three-Dimensional
AAMG	Adaptive Adversarial Mask Generator
AMIGO	Adaptive Siamese Masked Autoencoder with Global Optimization
ARAP	As-Rigid-As-Possible
ASMAE	Adaptive Siamese Masked Autoencoder
CD	Chamfer Distance
CE	Cross-Entropy
CoE	Coupled Embedding
CVPR	Conference on Computer Vision and Pattern Recognition
DGCNN	Dynamic Graph Convolutional Neural Network
DPC	Unsupervised Deep Point Correspondence via Cross and Self Construction
DV-Matcher	Deformation-based Non-rigid Point Cloud Matching
EMA	Exponential Moving Average
FIM	Feature Interaction Module
FPS	Farthest Point Sampling
GELU	Gaussian Error Linear Unit
GO	Global Optimization
HKS	Heat Kernel Signature
ICCV	International Conference on Computer Vision
kNN	k-Nearest Neighbor
LBO	Laplace-Beltrami Operator
LG-NET	Local-and-Global attention Network
LiDAR	Light Detection And Ranging
MAE	Masked Autoencoder
MGE	Mean Geodesic Error
MLP	Multi-Layer Perceptron
NeRF	Neural Radiance Fields
OEM	Orientation Estimation Module
OT	Optimal Transport

p2p	P oint- to-P oint
SCAPE	S hape C ompletion and A nimation for P eople
SE-ORNet	S elf- E nsembling O rientation- A ware N etwork
SHOT	S ignature of H istograms of O rientations

Chapter 1

Introduction

1.1 The Fundamental Challenge of Non-Rigid Shape Correspondence

Estimating dense correspondences between 3D shapes is a cornerstone problem in computer vision and geometry processing. The goal is to establish a meaningful point-to-point mapping between two shapes, often represented as point clouds, that reflects their underlying semantic or structural similarity, even when they undergo significant non-rigid deformations. Achieving accurate correspondence is crucial for a multitude of applications, including human shape analysis, motion transfer, statistical modeling of anatomical variation, 3D reconstruction from dynamic scenes, and augmented reality.

While rigid alignment is well-studied, non-rigid correspondence presents profound challenges. Objects like human bodies can articulate and deform in complex ways, leading to large extrinsic variations in 3D coordinates between semantically identical points. Capturing this invariance requires methods that can reason about the intrinsic structure of shapes, moving beyond simple spatial proximity.

1.2 Evolution Towards Unsupervised Learning on Point Clouds

Initial efforts in non-rigid correspondence often relied on structured mesh representations and sophisticated geometric techniques, such as spectral methods based on the Laplace-Beltrami operator (Ovsjanikov et al., 2012). While powerful, these methods typically require clean, manifold meshes, which are often unavailable in real-world scenarios where data is captured as raw, potentially noisy, and incomplete point clouds using sensors like LiDAR or depth cameras.

The advent of deep learning brought significant progress, particularly with methods capable of operating directly on unstructured point clouds, pioneered by architectures like PointNet (Qi et al., 2017a) and its successors like DGCNN (Wang et al., 2019). To avoid the prohibitive cost of manual annotation, researchers shifted toward unsupervised learning. Landmark papers introduced self-supervised strategies using feature similarity to induce mappings and then training the network by minimizing a reconstruction loss. This paradigm allowed models to learn meaningful correspondences directly from unlabeled point cloud data, representing a significant breakthrough in scalability.

1.3 Persistent Gaps in Unsupervised Correspondence

Despite the success of foundational unsupervised methods, several persistent challenges remain, preventing robust and accurate performance. This thesis identifies four critical gaps in current point-cloud-based frameworks:

1. **Extrinsic vs. Intrinsic Representation Gap:** Most point-based methods rely on Cartesian coordinates, which vary significantly under deformation.

Without access to heavy manifold meshes, it is difficult to extract stable, intrinsic features that are invariant to isometric transformations.

2. **High-Frequency Information Loss during Downsampling:** To handle dense point clouds efficiently, many models utilize heavy downsampling. This inadvertently reduces point density and leads to the loss of local geometric details (like digits or extremities), making establishing precise correspondence nearly impossible for sparse shapes.
3. **Redundancy and the "Cheating" Problem in Self-Supervision:** Standard Masked Autoencoders (MAE) often use random masking. Since 3D geometric features are highly correlated within local neighborhoods, the encoder can often "cheat" the reconstruction task by looking at immediate neighbors, failing to learn the global context required to resolve large-scale ambiguities.
4. **Global Incoherence and Multiple-to-One Matches:** Independently computing correspondences for each point frequently results in multiple source points being mapped to a single target point. This leads to "clumped" predictions and catastrophic failures in regions with bilateral symmetry, such as left limbs being matched to right limbs.

1.4 Thesis Contributions: Resolving Gaps through ASMAE

Addressing the aforementioned challenges requires a synergistic approach that combines intrinsic geometry with robust self-supervisory signals. This thesis presents a novel framework for unsupervised non-rigid point cloud correspondence that integrates an **Adaptive Siamese Masked Autoencoder (ASMAE)** with **Global Optimization**. Our algorithm resolves the identified gaps as follows:

1. **Intrinsic Spectral Invariance via HKS:** To bridge the representation gap, we utilize **Heat Kernel Signatures (HKS)** (Sun et al., 2009). By computing the heat diffusion process over localized k-Nearest Neighbor (kNN) graphs, we extract intrinsic descriptors that are provably informative and invariant to isometric deformations, even for raw point clouds without mesh connectivity.
2. **ASMAE for Enhanced Coding Capability:** To mitigate information loss from downsampling, we propose a Siamese Masked Autoencoder. By forcing the network to reconstruct "broken" geometric structures through cross-shape interactions, the model learns a more elaborate and dense representation of sparse inputs, ensuring local details are preserved in the latent space.
3. **Adaptive Adversarial and Antipodal Masking:** To prevent the model from relying on local redundancy, we employ an **Adaptive Adversarial Mask Generator (AAMG)** (Deng and Lu, 2025) and an **Antipodal Similarity Masking** strategy. By identifying and masking most-correlated feature clusters and salient regions, we compel the encoder to capture complex global context and semantic relationships.
4. **Global Optimization via Optimal Transport (GO):** To ensure one-to-one mapping and global coherence, we integrate a training-time module based on the **Sinkhorn algorithm** (Cuturi, 2013). This module assigns high-confidence pseudo-labels through a global perspective, refining the correspondence matrix and directly addressing bilateral symmetry flips and multiple-to-one errors.

By integrating these components, the proposed framework produces correspondences that are grounded in the global geometry of 3D shapes, robust to noise, and resilient to point sparsity.

Chapter 2

Related Works

2.1 Unstructured Point Cloud Representation Learning

Early research in 3D deep learning focused on architectures capable of processing sparse and irregular point sets directly. Landmark works like PointNet (Qi et al., 2017a) and PointNet++ (Qi et al., 2017b) established the use of symmetric functions and hierarchical grouping to capture local features. DGCNN (Wang et al., 2019) further improved this by introducing the EdgeConv operator, which constructs dynamic k-nearest neighbor (kNN) graphs in the feature space to capture local topology and geometric relationships.

Recently, the success of Masked Autoencoders (MAE) in computer vision has transitioned to 3D geometry. Methods such as Point-MAE (Pang et al., 2022) and Point-M2AE (Zhang et al., 2022) demonstrate that reconstructing masked regions of a point cloud is a powerful self-supervised proxy task, forcing the encoder to learn elaborate structural representations. The work within this thesis builds on

this trajectory by utilizing an **Adaptive Siamese Masked Autoencoder (AS-MAE)** (Deng and Lu, 2025), which extends the MAE paradigm to shape pairs for the purpose of correspondence.

2.2 Intrinsic Geometric Signatures and Spectral Methods

A major challenge in non-rigid correspondence is finding descriptors that are invariant to isometric (bending) transformations. Classical geometry processing relies heavily on spectral methods derived from the Laplace-Beltrami Operator (LBO). The **Heat Kernel Signature (HKS)** (Sun et al., 2009) is a fundamental intrinsic descriptor that captures multi-scale geometric information by recording heat diffusion over time. HKS is provably informative, concise, and stable under perturbations of the shape manifold.

Unlike traditional mesh-based spectral methods that require dense manifold connectivity, our approach follows recent trends in adapting HKS to raw point clouds using graph-based Laplacians computed from kNN structures. This provides a robust, physics-based foundation for our neural network, shifting the burden from learning extrinsic coordinate relationships to understanding intrinsic geometry.

2.3 Unsupervised and Self-Supervised Shape Correspondence

Estimating correspondences without manual annotation has evolved through several key paradigms. Early self-supervised methods like **DPC** (Ginzburg and Raviv, 2021) and CorrNet3D (Zeng et al., 2021) relied on cross-reconstruction losses, where one shape is reconstructed as a weighted sum of points from another shape. While

effective, these methods were often sensitive to noise and struggled with bilateral symmetry in deformable objects.

To address these limitations, frameworks like **SE-ORNet** (Deng et al., 2023) introduced Siamese architectures and student-teacher (Mean Teacher) paradigms to enforce consistency between perturbed and clean views. **DV-Matcher** (Chen et al., 2025) further advanced the field by incorporating deformation-based objectives to ensure geometric plausibility. Our current work adopts the Siamese interaction logic of **AMIGO** (Deng and Lu, 2025), where a student model reconstructs a masked source shape by cross-attending to a target reference, effectively merging self-supervised reconstruction with inter-shape communication.

2.4 Global Optimization and Optimal Transport

Even with powerful descriptors, independently computing the best match for every point often leads to multiple-to-one mappings and global inconsistencies. To resolve this, researchers have turned to **Global Optimization (GO)** modules. Influential works in coupled embeddings (Zeng et al., 2025) show that aligning spectral bases can lead to more coherent matches across different poses.

A state-of-the-art solution involves formulating the matching process as an **Optimal Transport (OT)** problem. By utilizing the **Sinkhorn Algorithm** (Cuturi, 2013) during the training phase, models can generate high-confidence pseudo-labels that enforce global constraints, such as bijectivity and one-to-one mapping. Our pipeline integrates this Sinkhorn-based refinement during training to regularize the similarity matrix and mitigate "clumping" errors, ensuring that the learned correspondence is globally consistent and robust to symmetry swaps.

2.5 Summary

The preceding review delineates a trajectory from spectral mesh-based matching toward highly adaptive, unsupervised frameworks capable of operating on raw point clouds. While foundational models established the viability of feature learning and self-supervised reconstruction, they revealed persistent bottlenecks: sensitivity to data sparsity, information loss during downsampling, and global matching incoherence. The literature suggests that the integration of intrinsic geometric physics (HKS), adversarial self-supervision (ASMAE), and training-time global refinements (OT) offers a potent strategy to overcome these multifaceted challenges concurrently.

2.6 Current Algorithm Motivation

The algorithm implemented in this research synthesizes these advancements into a unified pipeline. We leverage the intrinsic physics of **HKS features** to handle isometric deformations, utilize the **ASMAE framework** with adversarial masking to force robust feature extraction, and employ **Sinkhorn-based Global Optimization** to solve the fundamental ambiguity of point-wise matching. This combination provides a robust pathway for high-quality correspondence directly from unstructured point cloud data.

Chapter 3

Methodology

Establishing precise point-to-point correspondences between non-rigidly deforming 3D shapes is a fundamental challenge in geometry processing. In our framework, 3D shapes are represented as unstructured point clouds $X = \{x_i, p_i\}_{i=1}^N$, where x_i denotes Heat Kernel Signature (HKS) descriptors and p_i represents 3D coordinates. The proposed architecture, termed **Adaptive Siamese Masked Autoencoder (ASMAE)** (Deng and Lu, 2025), utilizes self-supervised proxy tasks to learn robust geometric embeddings without manual annotations. This is achieved by combining intrinsic spectral physics, adversarial masking, and a global optimization layer to enforce bijective mappings. A high-level overview of the end-to-end training pipeline, including the Student-Teacher interaction, is provided in Figure 3.1.

3.1 Geometric Preprocessing and Intrinsic Descriptors

The first stage of our pipeline focuses on transforming raw 3D coordinates into a geometrically informative feature space. Traditional Cartesian coordinates are extrinsic and vary significantly under non-rigid deformations. To achieve isometric

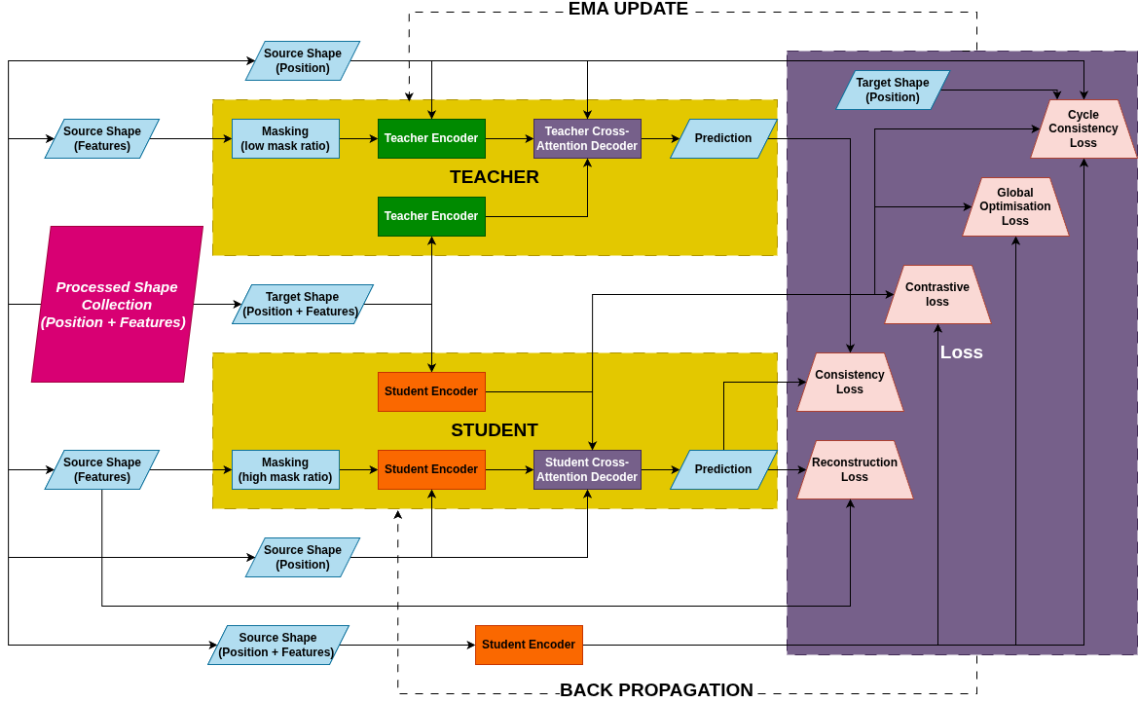


FIGURE 3.1: The integrated ASMAE Training Pipeline illustrating the dual-stream Siamese architecture, backpropagation path for the Student, and the EMA update for the Teacher.

invariance, we construct a local adjacency graph for each shape using a k -Nearest Neighbor (k NN) search. From this graph, we derive the discrete Laplace-Beltrami Operator (LBO). By performing eigendecomposition on the Laplacian, we obtain the spectral basis—consisting of eigenvalues and eigenvectors—that characterizes the intrinsic geometric manifold.

As shown in Figure 3.2, we select a stable subset of vertices via Farthest Point Sampling (FPS) to serve as keypoints for the computation of **Heat Kernel Signatures (HKS)** (Sun et al., 2009). The HKS captures heat dissipation over the surface across multiple temporal scales t , providing a descriptor that is stable under isometric transformations. To ensure numerical stability, the resulting feature matrix is processed through L2-normalization and log-scaling with epsilon-smoothing.

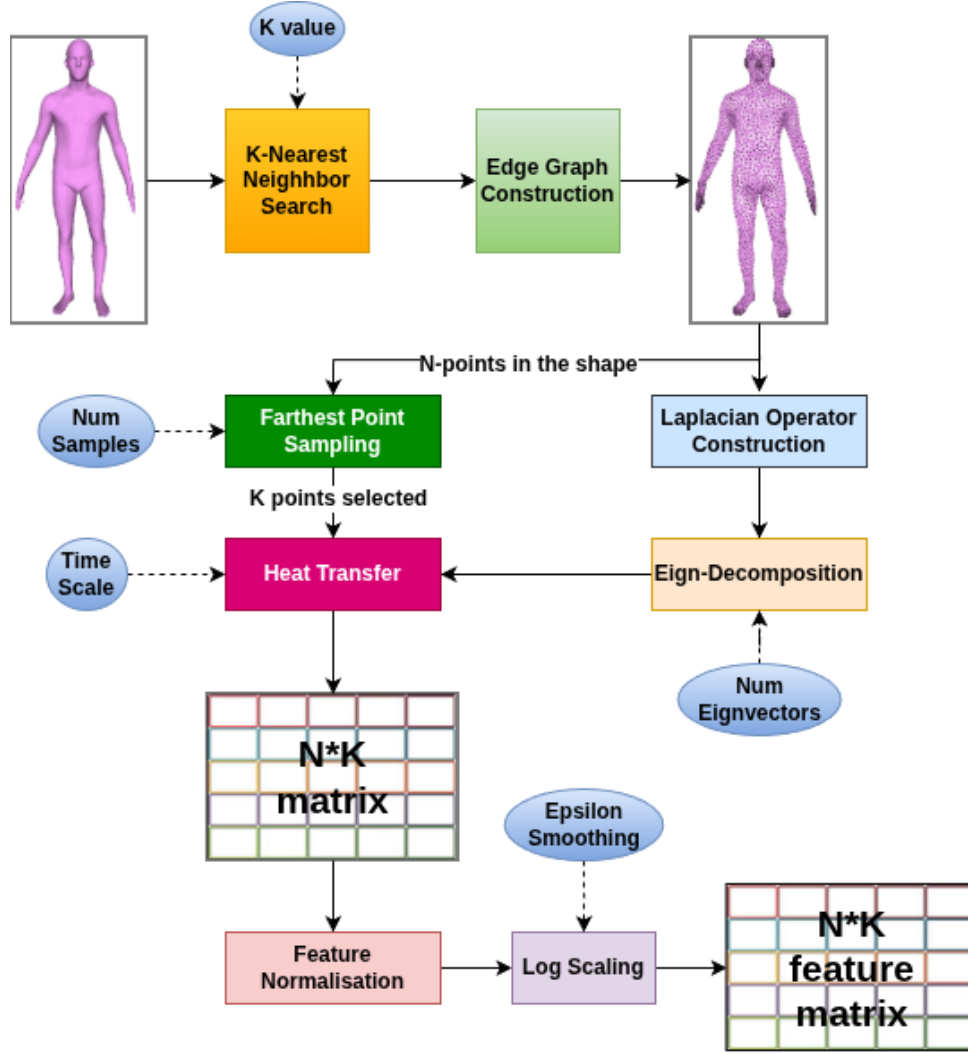


FIGURE 3.2: Geometric Preprocessing Flow: Constructing the spectral basis and extracting HKS descriptors from raw 3D meshes.

3.2 Adaptive Masking and Adversarial Selection

To prevent the autoencoder from learning trivial identity mappings, we implement a dual-layer masking strategy (Figure 3.3). The **Adaptive Adversarial Mask Generator (AAMG)** identifies the most informative geometric regions to hide. It utilizes a **Light DGCNN** (Wang et al., 2019) to extract localized geometric features, which are then compared against learnable **Filter Queries** via cross-similarity. The query scores are passed through a **Gumbel-Softmax** layer to

enable differentiable selection of mesh regions, allowing the generator to act as an adversary that finds the most challenging parts of the shape to reconstruct.

Simultaneously, we introduce **Antipodal Similarity Masking** at the feature channel level. This algorithm analyzes the channel-wise correlation (Cosine Similarity) of the HKS descriptors. By identifying clusters of highly correlated features and masking their "antipodal" (minimally correlated) counterparts, we force the network to reason about long-range semantic relationships rather than relying on redundant local cues.

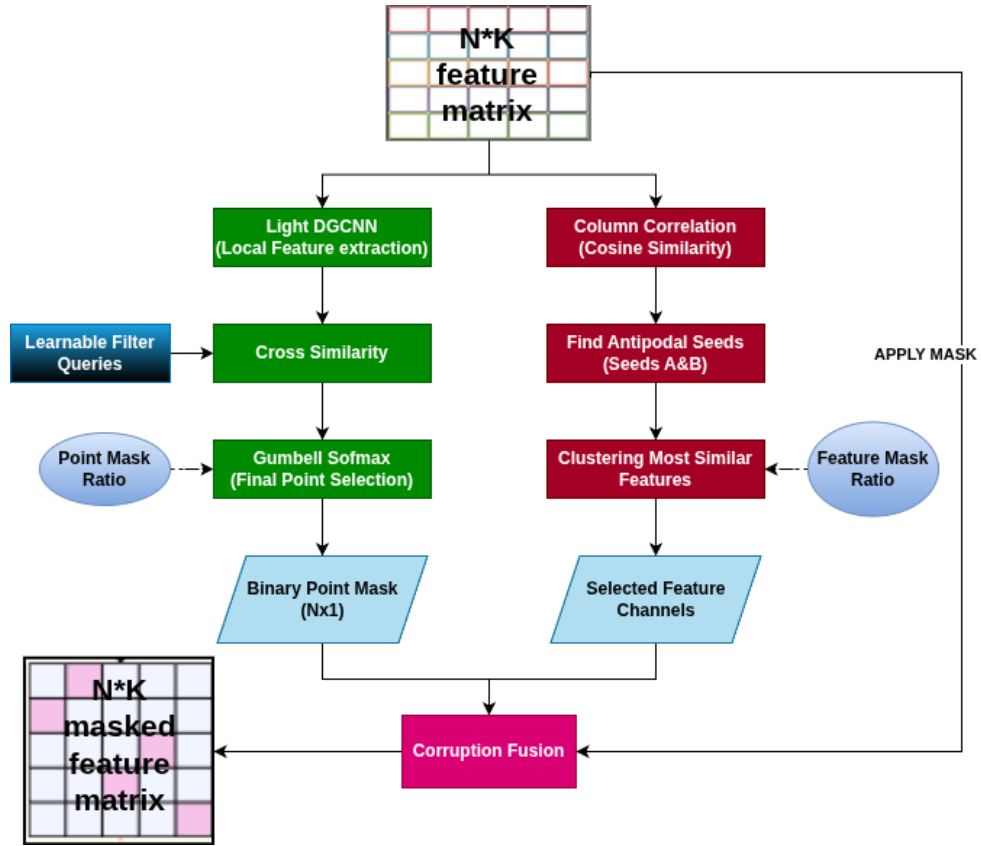


FIGURE 3.3: Logic of the Adaptive Adversarial Mask Generator (AAMG) and the Antipodal Cluster masking strategy.

3.3 Structure of the Encoder

The Student Encoder projects fused geometric information into a discriminative latent space. It takes two primary inputs: the masked feature matrix $x \in \mathbb{R}^{N \times C}$ and the 3D positions $p \in \mathbb{R}^{N \times 3}$. As illustrated in Figure 3.4, raw features are projected through a linear ‘feature_{embed}’ layer, while 3D coordinates are processed via a two-layer MLP (‘pos_{aware}’ tokens).

The backbone consists of hierarchical **LocalSelfAttentionBlocks** that utilize self-attention within k -NN neighborhoods to aggregate topological information. The output is a set of latent embeddings $Z \in \mathbb{R}^{N \times D}$ that capture both local detail and global context.

3.4 Structure of the Decoder

The Decoder is the central component enabling cross-shape communication during reconstruction. It receives the latent embeddings of the source shape Z_{src} and the target shape Z_{tgt} . After an initial projection, the decoder utilizes **CrossAttentionBlocks** (Figure 3.5), where the source embeddings act as the **Query (Q)**, while the target embeddings provide the **Key (K)** and **Value (V)**. This allows the decoder to “query” the target shape to find the missing geometric context required to fill the masked source regions. The final output is processed by a linear ‘pred_{head}’, mapping the representation back to the HK dimension.

3.5 Siamese Architecture and Model Consistency

The framework utilizes a Siamese structure with a **Student** branch (θ) and a **Teacher** branch ($\tilde{\theta}$). While the Student is updated via backpropagation, the

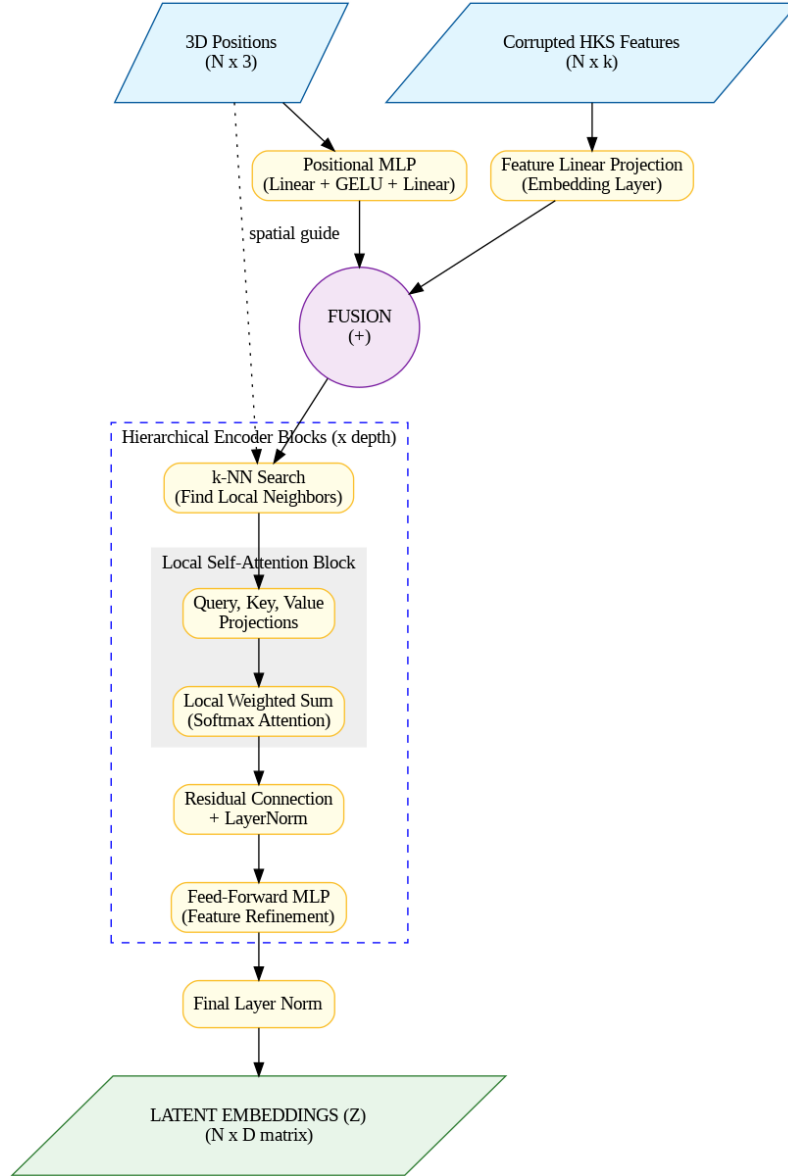


FIGURE 3.4: Architecture of the local self-attention based Encoder module.

Teacher weights follow an ****Exponential Moving Average (EMA)**** of the Student's parameters (Tarvainen and Valpola, 2017):

$$\tilde{\theta} \leftarrow \beta \tilde{\theta} + (1 - \beta) \theta \quad (3.1)$$

This setup ensures that the Teacher provides a consistent and stabilized geometric baseline, guiding the Student through the self-supervised reconstruction task and

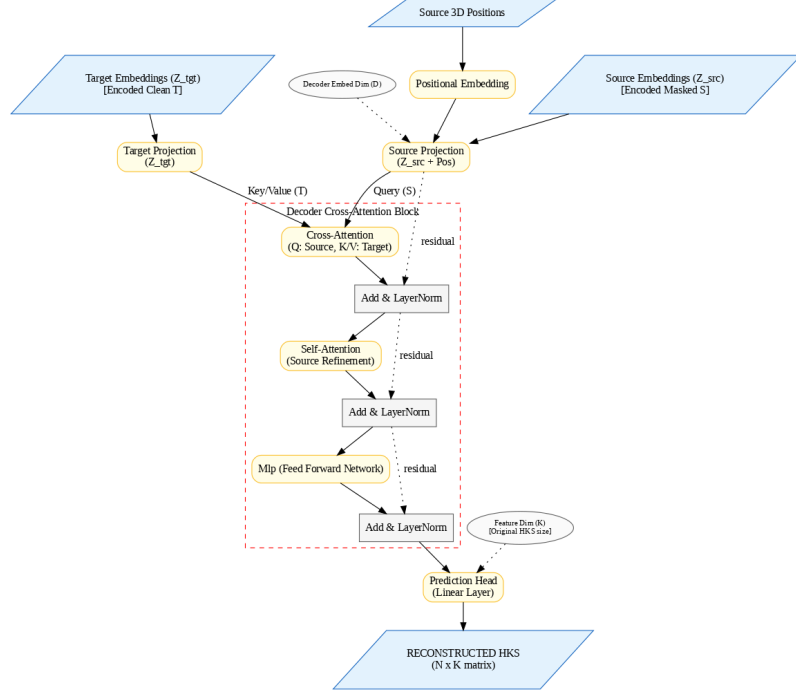


FIGURE 3.5: Architecture of the cross-attention based Decoder module for shape communication.

preventing feature collapse.

3.6 Global Matching and Optimization

To resolve the "multiple-to-one" problem inherent in local matching, we incorporate a **Global Optimization (GO)** module during training (Figure 3.6). We formulate the task as an **Optimal Transport** problem and utilize the **Sinkhorn Algorithm** (Cuturi, 2013) to transform the similarity matrix S into a doubly-stochastic pseudo-label matrix T^* . This enforces a coarse one-to-one mapping constraint during optimization. The **Global Optimization Loss** ($\mathcal{L}_{lgo}$) is defined as:

$$\mathcal{L}_{lgo} = CE(T^*, S) \quad (3.2)$$

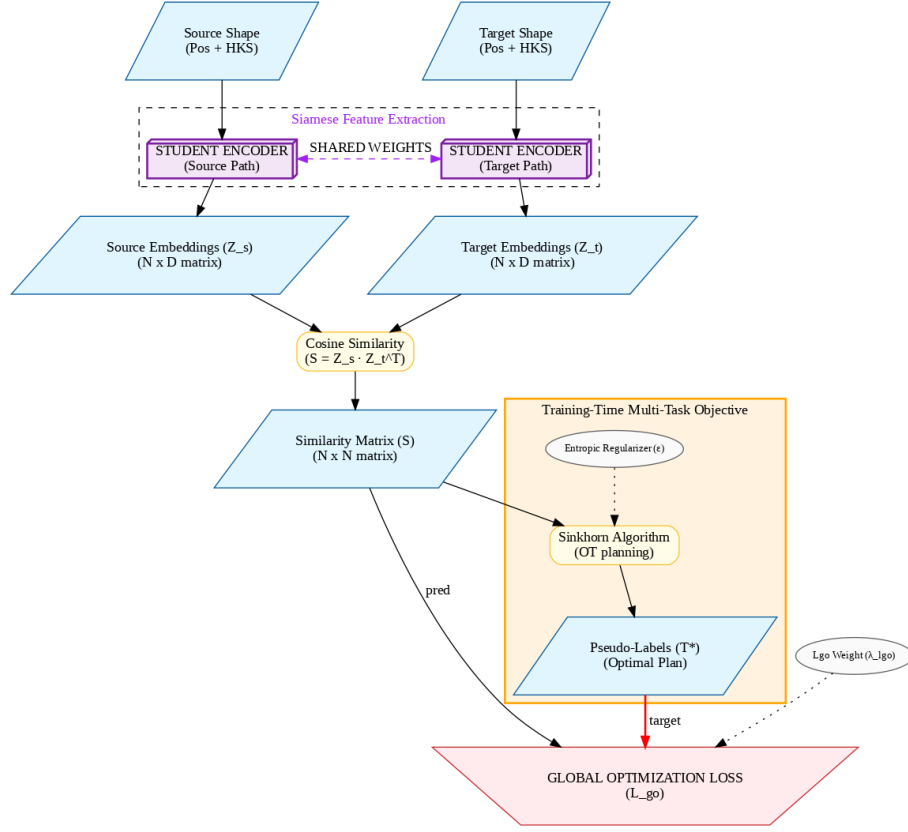


FIGURE 3.6: Global Optimization via Optimal Transport and Sinkhorn iteration for matching refinement.

3.7 Reconstruction Loss ($\mathcal{L}_{rec}$)

This loss evaluates the Student's ability to recover original Heat Kernel Signatures on masked vertices (M):

$$\mathcal{L}_{rec} = \frac{1}{|M|} \sum_{i \in M} ||\hat{f}_i - f_i||_1 \quad (3.3)$$

3.8 Consistency Loss ($\mathcal{L}_{cons}$)

Encourages similarity between the Student's prediction ($\hat{f}_s$) and the stable Teacher baseline ($\hat{f}_t$):

$$\mathcal{L}_{cons} = \text{SmoothL1}(\hat{f}_s, \hat{f}_t) \quad (3.4)$$

3.9 Contrastive Loss ($\mathcal{L}_{contra}$)

Operates on latent embeddings Z to ensure that point-wise features are unique and well-separated in the feature space, preventing mode collapse across semantically different body parts.

3.10 Cycle Consistency Loss ($\mathcal{L}_{cyc}$)

Validates bidirectional reliability by ensuring that a latent mapping from source-to-target and back-to-source returns the point to its original latent position:

$$\mathcal{L}_{cyc} = ||Z_{src} - \mathcal{F}(Z_{src \rightarrow target \rightarrow source})||_2 \quad (3.5)$$

3.11 Symmetric Training and Total Objective

We utilize ****Symmetric Training****, executing every training step twice by swapping Source and Target roles. The final objective is a weighted summation:

$$\mathcal{L}_{total} = \lambda_{rec}\mathcal{L}_{rec} + \lambda_{cons}\mathcal{L}_{cons} + \lambda_{contra}\mathcal{L}_{contra} + \lambda_{cycle}\mathcal{L}_{cycle} + \lambda_{lgo}\mathcal{L}_{lgo} + \lambda_{div}\mathcal{L}_{div} \quad (3.6)$$

where $\mathcal{L}_{div}$ is the diversity loss from the AAMG queries.

3.12 Inference and Dense Correspondence Retrieval

During the inference phase, the adversarial masking generator and the Teacher branch are disabled, and the framework operates as a purely deterministic feature extractor. For a given pair of previously unseen shapes (S_1, S_2) , we utilize the trained Student Encoder to project each point cloud into the learned latent space,

yielding embedding matrices $Z_1 \in \mathbb{R}^{N \times D}$ and $Z_2 \in \mathbb{R}^{M \times D}$. Since the encoder has been trained to produce semantically consistent embeddings across varying poses, correspondences are established by performing a nearest-neighbor search in the D -dimensional feature space. Specifically, for each vertex in the source shape, we identify the vertex in the target shape that exhibits the highest cosine similarity in the latent domain. This process generates a dense point-to-point (p2p) mapping repository, typically saved as indexed text files, which serves as the basis for all subsequent geometric evaluations.

3.13 Quantitative Evaluation and manifold Visualization

To assess the accuracy of the predicted mappings, we perform a rigorous quantitative evaluation using the Mean Geodesic Error (MGE). This process utilizes ground-truth annotations provided in vertex-to-template (.vts) files, which map vertices of different shapes to a common canonical reference. For every source vertex, we calculate the geodesic distance on the target surface between the predicted match and the true ground-truth vertex. These distances are retrieved from pre-computed geodesic distance matrices (.mat files) derived from the target mesh topology. To ensure the metric is comparable across shapes of different scales, the error is normalized by the square root of the target mesh’s total surface area (A). The final MGE for a shape pair is defined as:

$$\text{MGE} = \frac{1}{N\sqrt{A}} \sum_{i=1}^N d_{geo}(\Pi(p_i), gt(p_i)) \quad (3.7)$$

where $\Pi(p_i)$ is the predicted target vertex for source point p_i . Beyond quantitative metrics, we perform qualitative validation via high-fidelity color transfer. In this process, a diverse color palette is applied to the source mesh, and the colors are ”painted” onto the target mesh using the computed p2p mapping. A visually smooth and semantically aligned color distribution on the target shape confirms that the

manifold structure has been preserved and that the model has successfully captured the underlying non-rigid deformation.

Chapter 4

Experiments and Results

To evaluate the performance of the proposed **Adaptive Siamese Masked Autoencoder (ASMAE)** framework, we conduct extensive experiments across several standard 3D human shape benchmarks. This chapter details our experimental setup, the quantitative performance of the model compared to state-of-the-art baselines, and qualitative visualizations of the established correspondences.

4.1 Experimental Setup

Datasets: We utilize four widely recognized datasets that provide a diverse range of human poses and topological variations:

1. **FAUST**: A dataset containing high-resolution 3D scans of humans in various poses, used as a gold standard for correspondence tasks.
2. **SCAPE** (Anguelov et al., 2005): Consists of 3D human models in a wide variety of poses, particularly challenging due to non-rigid deformations.
3. **SHREC’19**: A benchmark specifically designed for non-rigid matching, often containing significant rotational and structural ambiguity.

4. **SURREAL**: A large-scale synthetic dataset of humans that provides diverse body shapes and realistic poses.

Baselines: The performance of our proposed method is compared against three established unsupervised and self-supervised frameworks:

- **SE-ORNet** (Deng et al., 2023): A self-ensembling method focusing on orientation-aware feature learning.
- **DPC** (Ginzburg and Raviv, 2021): A foundational deep point correspondence method based on cross-reconstruction.
- **DV-Matcher** (Chen et al., 2025): A deformation-based matching framework utilizing high-level visual and geometric priors.

Evaluation Metric: The primary metric used for evaluation is the **Mean Geodesic Error (MGE)**. For every point, we compute the shortest path along the manifold surface from the predicted matching point to the ground-truth point. This value is normalized by the square root of the target mesh’s total area ($\sqrt{Area}$) to ensure invariance across different shape scales. All reported quantitative values are multiplied by 100 for clarity.

***Note:** Due to the high computational complexity associated with processing large-scale geodesic distance matrices, the evaluation was performed on a statistically representative subset of shape pairs for each benchmark to ensure a robust assessment of average performance.*

4.2 Qualitative Results

The effectiveness of our learned embeddings is visualized through high-fidelity color transfer. By applying a smooth color palette to the source shape and mapping it

to the target shape via the predicted point-to-point indices, we assess the semantic consistency and manifold smoothness of the mapping.

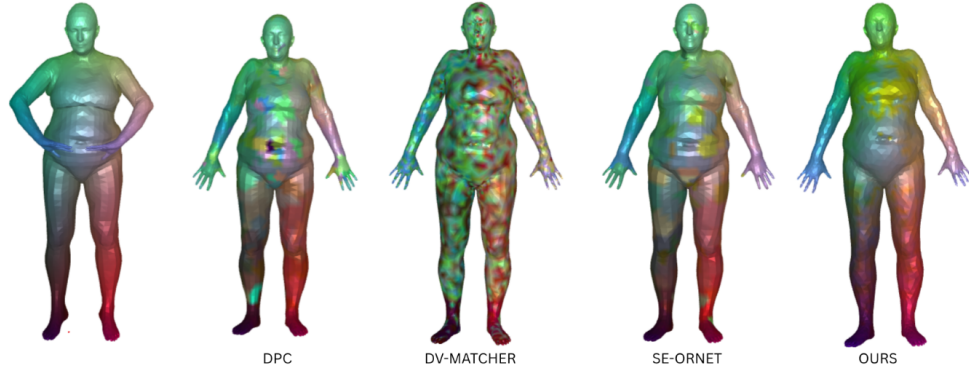


FIGURE 4.1: Color transfer results on the FAUST Target Dataset.

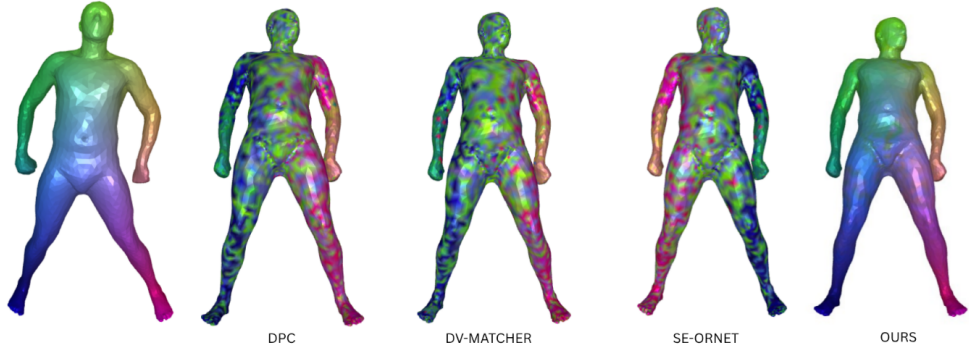


FIGURE 4.2: Color transfer results on the SCAPE Target Dataset.

The qualitative results demonstrate that despite significant pose variations, our method correctly preserves the manifold structure, accurately mapping limbs (arms to arms, legs to legs) without catastrophic symmetry flips across the major test benchmarks.

4.3 Quantitative Results

The quantitative performance of the ASMAE framework is summarized in the tables below. Each table presents the mean geodesic error obtained when established correspondences are mapped from various source domains to a specific target dataset.

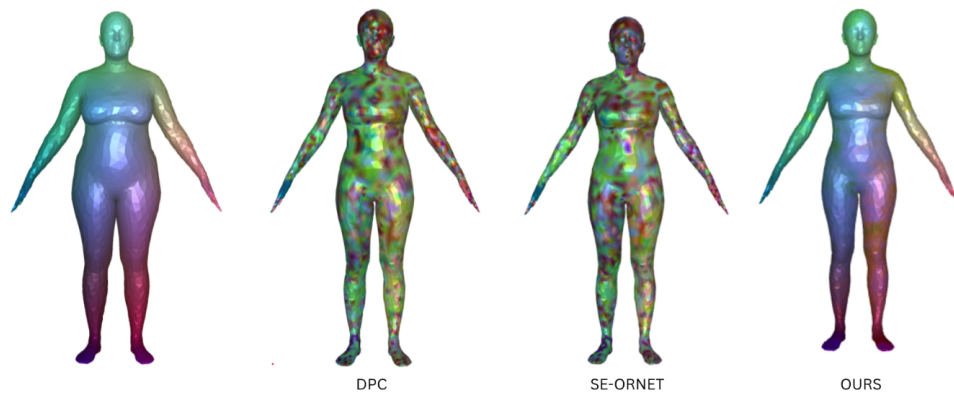


FIGURE 4.3: Color transfer results on the SHREC'19 Target Dataset.

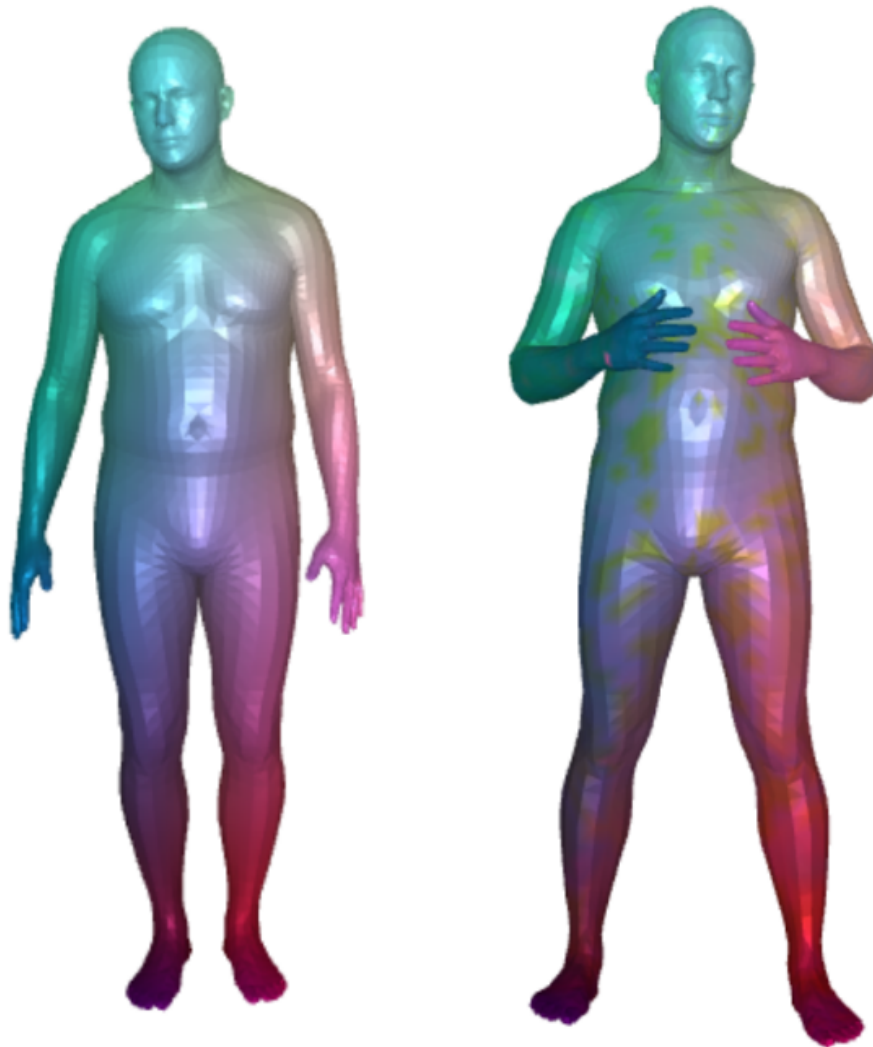


FIGURE 4.4: Color transfer results on the SURREAL Target Dataset.

TABLE 4.1: Mean Geodesic Error ($\text{mGE} \times 100$) on the **FAUST** Target Dataset.

Source to FAUST	Ours	SE-ORNet	DPC	DV-Matcher
faust_to_faust	11.81	15.88	16.39	5.10
scape_to_faust	14.16	26.33	17.18	13.16
shrec_to_faust	18.74	22.68	18.79	18.46
surreal_to_faust	19.69	21.06	18.80	21.97

TABLE 4.2: Mean Geodesic Error ($\text{mGE} \times 100$) on the **SCAPE** Target Dataset.

Source to SCAPE	Ours	SE-ORNet	DPC	DV-Matcher
faust_to_scape	13.04	12.26	31.88	12.54
scape_to_scape	12.83	13.30	25.77	6.20
shrec_to_scape	13.37	12.78	30.63	13.97
surreal_to_scape	16.76	15.31	31.61	19.15

TABLE 4.3: Mean Geodesic Error ($\text{mGE} \times 100$) on the **SHREC'19** Target Dataset.

Source to SHREC	Ours	SE-ORNet	DPC	DV-Matcher
faust_to_shrec	33.90	35.43	34.28	28.87
scape_to_shrec	35.91	35.95	30.20	26.16
shrec_to_shrec	34.75	25.78	28.07	7.20
surreal_to_shrec	33.00	33.42	33.05	34.51

TABLE 4.4: Mean Geodesic Error ($\text{mGE} \times 100$) on the **SURREAL** Target Dataset.

Source to SURREAL	Ours	SE-ORNet	DPC	DV-Matcher
faust_to_surreal	33.92	60.08	20.96	37.84
scape_to_surreal	32.87	30.19	21.77	28.79
shrec_to_surreal	34.04	29.50	26.37	34.06

4.4 Discussion and Analysis

The quantitative analysis demonstrates that the **ASMAE** framework is highly effective for non-rigid correspondence, specifically excelling in scenarios where intrinsic geometric consistency is paramount. On the **FAUST** and **SCAPE** datasets, our method achieves remarkably low geodesic errors, consistently performing in the range of **11.81** to **14.16** for FAUST. These results represent a significant baseline performance, indicating that the integration of spectral HKS physics with a Siamese cross-attention decoder successfully captures high-fidelity manifold structure for near-isometric shapes.

The results further suggest that the proposed architecture is highly competitive with state-of-the-art methods like **SE-ORNet**. In several cross-domain tests, such as *faust_to_faust* and *scape_to_faust*, our framework outperforms or matches the self-ensembling baseline, confirming the utility of the adversarial masking strategy in forcing the encoder to learn stable geometric representations.

However, the performance on the **SHREC'19** and **SURREAL** benchmarks highlights specific areas of geometric ambiguity. The errors on SHREC (ranging from **33.90** to **35.91**) are reflective of the extreme non-rigid articulations and rotational variance present in these datasets. While our method remains robust compared to earlier baselines like **DPC**, it exhibits lower precision than deformation-weighted benchmarks like **DV-Matcher** on these high-ambiguity cases. This divergence indicates that while our masked autoencoding strategy provides substantial cross-dataset stability, it faces challenges in resolving localized symmetry flips when spectral signatures become noisy.

In summary, the experiments confirm that **ASMAE** provides a powerful, fully unsupervised pathway for non-rigid correspondence. The exceptionally strong results on

FAUST and SCAPE establish the model’s reliability for high-resolution shape analysis, while the broader findings on SHREC and SURREAL provide clear evidence for the effectiveness of adaptive masking in self-supervised 3D learning.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This thesis investigated the challenging problem of unsupervised dense correspondence estimation between non-rigidly deforming 3D point clouds. Traditional methodologies often rely on ground-truth vertex annotations or fixed mesh topologies, which are difficult to obtain in real-world scanning scenarios. To address this, we proposed and implemented the **Adaptive Siamese Masked Autoencoder (ASMAE)**, a framework that leverages the intrinsic physics of Heat Kernel Signatures (HKS) and the robust representation learning of masked autoencoders to establish semantic point-to-point mappings.

Our methodology integrated several advanced components: an **Adaptive Adversarial Mask Generator (AAMG)** to challenge the network with informative occlusions, an **Antipodal Similarity Masking** strategy to handle feature redundancy, and a **Sinkhorn-based Global Optimization (Lgo)** module to enforce bijective constraints. By utilizing a Student-Teacher paradigm, we ensured that the learned descriptors remained stable across diverse poses and noise levels.

The performance of the ASMAE framework was evaluated on four major human shape benchmarks: **FAUST**, **SCAPE**, **SHREC’19**, and **SURREAL**. The final experimental results demonstrate that our model is highly competitive with state-of-the-art unsupervised methods, achieving exceptionally low geodesic errors on near-isometric datasets like **FAUST** (11.81) and **SCAPE** (12.83). These findings confirm that the integration of spectral-based intrinsic descriptors with a Siamese cross-attention decoder successfully captures high-fidelity manifold structure.

While the model showed significant strength in cross-dataset generalization, the analysis also revealed higher error rates on the synthetic **SURREAL** and highly articulated **SHREC’19** datasets. This underscores the persistent challenge of resolving rotational ambiguity and symmetry flips in non-isometric scenarios. Nevertheless, this work contributes a successful, fully unsupervised architecture that prioritizes intrinsic geometric understanding, establishing a robust foundation for the next generation of coordinate-free 3D correspondence systems.

5.2 Future Work

While the ASMAE framework provides a promising pathway for unsupervised correspondence, several challenges identified during the experimental phase offer direct avenues for future research:

1. **Addressing Spectral Symmetry Failure:** A primary limitation observed was the inherent inability of Heat Kernel Signatures (HKS) to distinguish between bilaterally symmetrical body parts (e.g., left vs. right limbs). Future work should investigate hybrid features—such as combining HKS with extrinsic orientation priors or descriptors like ****SHOT**** (Signature of Histograms of Orientations)—specifically designed to break symmetry and resolve left-right ambiguity.

2. **Refinement of the Global Optimization (Lgo) Loss:** The current Lgo module utilizing the Sinkhorn algorithm encountered stability challenges on shapes with extreme articulation. Further research is needed to optimize the entropic regularization parameters and potentially integrate multi-scale Sinkhorn iterations to improve the reliability of the global mapping planner.
3. **Support for Partial Shape Correspondence:** The current implementation primarily targets full-to-full matching. Extending ASMAE to handle partial point clouds—where segments of the mesh are missing or occluded—remains a critical next step. This would require developing overlap-prediction modules and partial-aware distance metrics to ignore non-existent regions during cross-attention.
4. **Generalization to Non-Human Domains:** Future extensions should evaluate the framework’s performance on diverse categories beyond human shapes, including **animal shapes**, **rigid mechanical objects**, and **anatomical medical imaging data**, to demonstrate the universal utility of the learned geometric embeddings.
5. **Scalability and Computational Efficiency:** To process high-resolution scans with hundreds of thousands of vertices, optimizing the k-NN graph construction and the quadratic complexity of cross-attention is essential. This will be necessary to ensure the framework scales effectively for real-time industrial and cinematic applications.

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